

Deterministic Optimization

Discrete Optimization:
Branch-and-Bound

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An Example

Example of the Branch-and-Bound Algorithm

Learning Objectives

- Observe an example of the branch-and-bound algorithm

Example

We would like to solve:

$$\begin{aligned} v^{IP} = \min \quad & 3x_1 + 2x_2 + x_3 \\ \text{s.t.} \quad & 3x_1 + 4x_2 + 5x_3 \geq 6 \\ & x_1, x_2, x_3 \geq 0 \text{ and integer} \end{aligned}$$

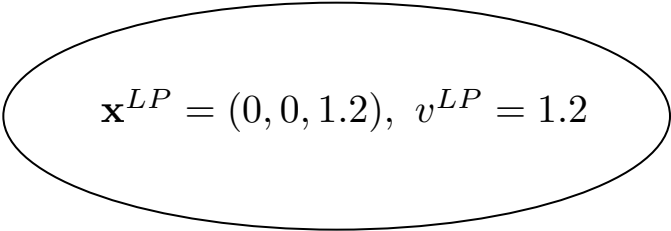
The LP relaxation:

$$\begin{aligned} v^{LP} = \min \quad & 3x_1 + 2x_2 + x_3 \\ \text{s.t.} \quad & 3x_1 + 4x_2 + 5x_3 \geq 6 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Branch-and-Bound

$$LB = 1.2 \leq v^{IP} \leq UB = +\infty$$

$$\begin{aligned} v^{LP} = \min \quad & 3x_1 + 2x_2 + x_3 \\ \text{s.t.} \quad & 3x_1 + 4x_2 + 5x_3 \geq 6 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$


$$\mathbf{x}^{LP} = (0, 0, 1.2), \quad v^{LP} = 1.2$$

Branch-and-Bound

$$LB = 1.2 \leq v^{IP} \leq UB = +\infty$$

$$\begin{aligned} v^{LP} = \min \quad & 3x_1 + 2x_2 + x_3 \\ \text{s.t.} \quad & 3x_1 + 4x_2 + 5x_3 \geq 6 \\ & x_3 \leq 1 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

$\mathbf{x}^{LP} = (0, 0, 1.2), v^{LP} = 1.2$

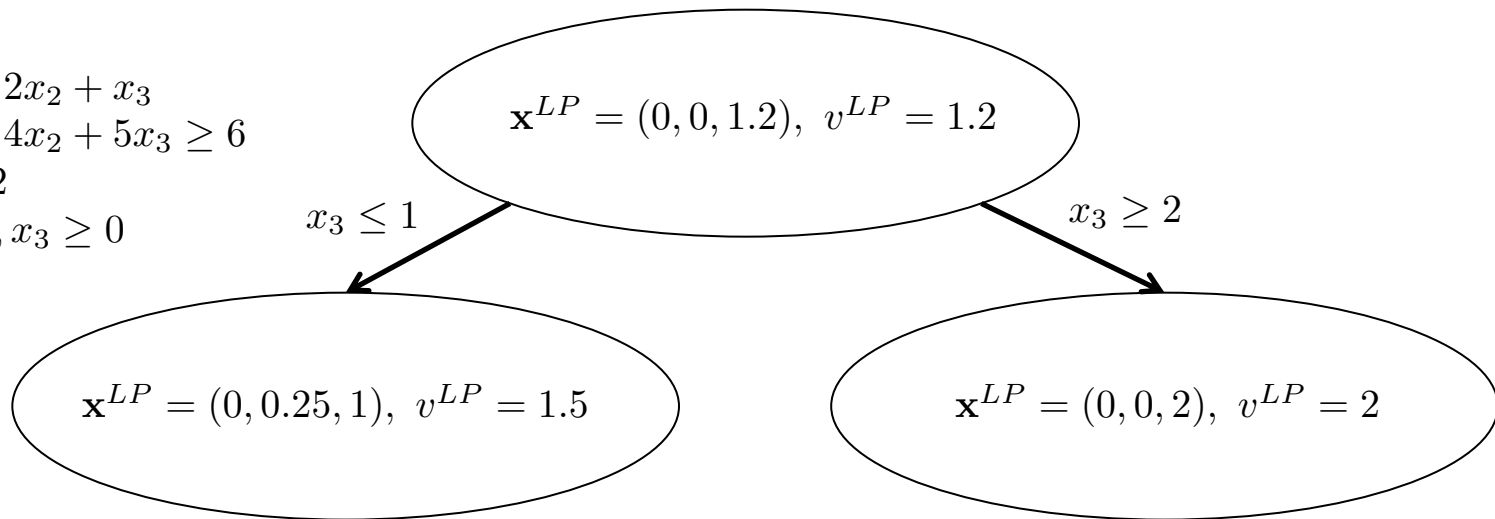
$x_3 \leq 1$

$\mathbf{x}^{LP} = (0, 0.25, 1), v^{LP} = 1.5$

Branch-and-Bound

$$LB = 1.5 \leq v^{IP} \leq UB = 2$$

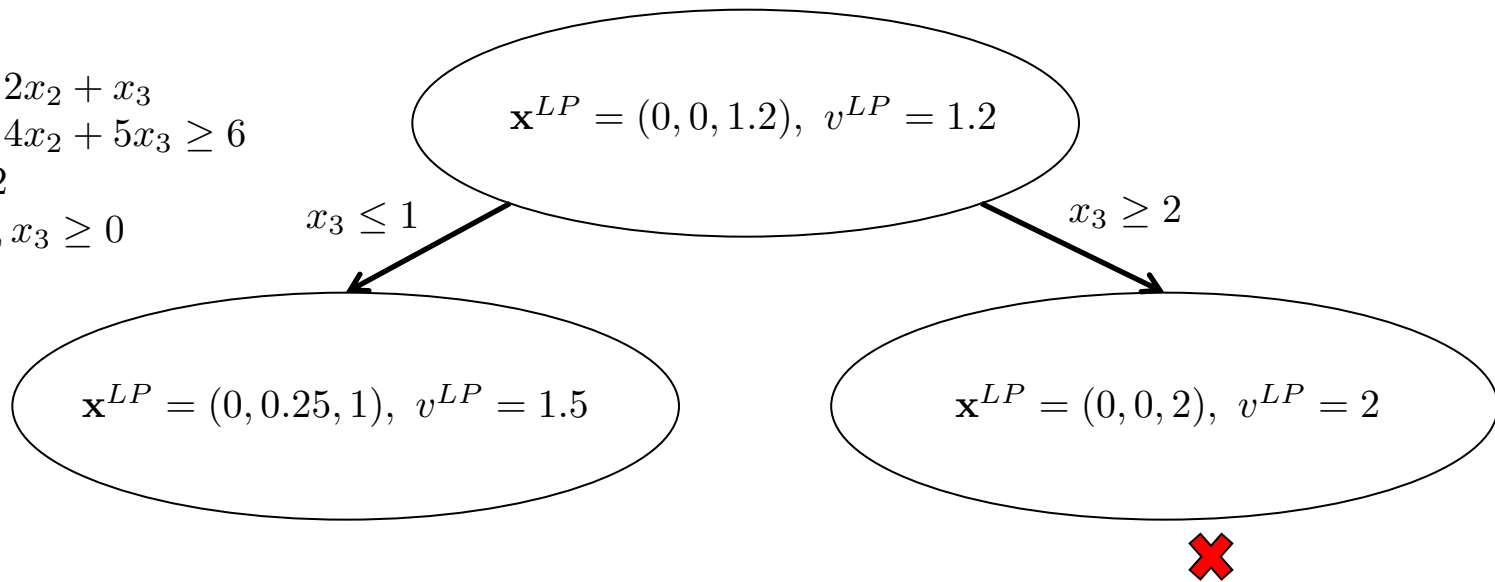
$$\begin{aligned} v^{LP} = \min \quad & 3x_1 + 2x_2 + x_3 \\ \text{s.t.} \quad & 3x_1 + 4x_2 + 5x_3 \geq 6 \\ & x_3 \geq 2 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$



Branch-and-Bound

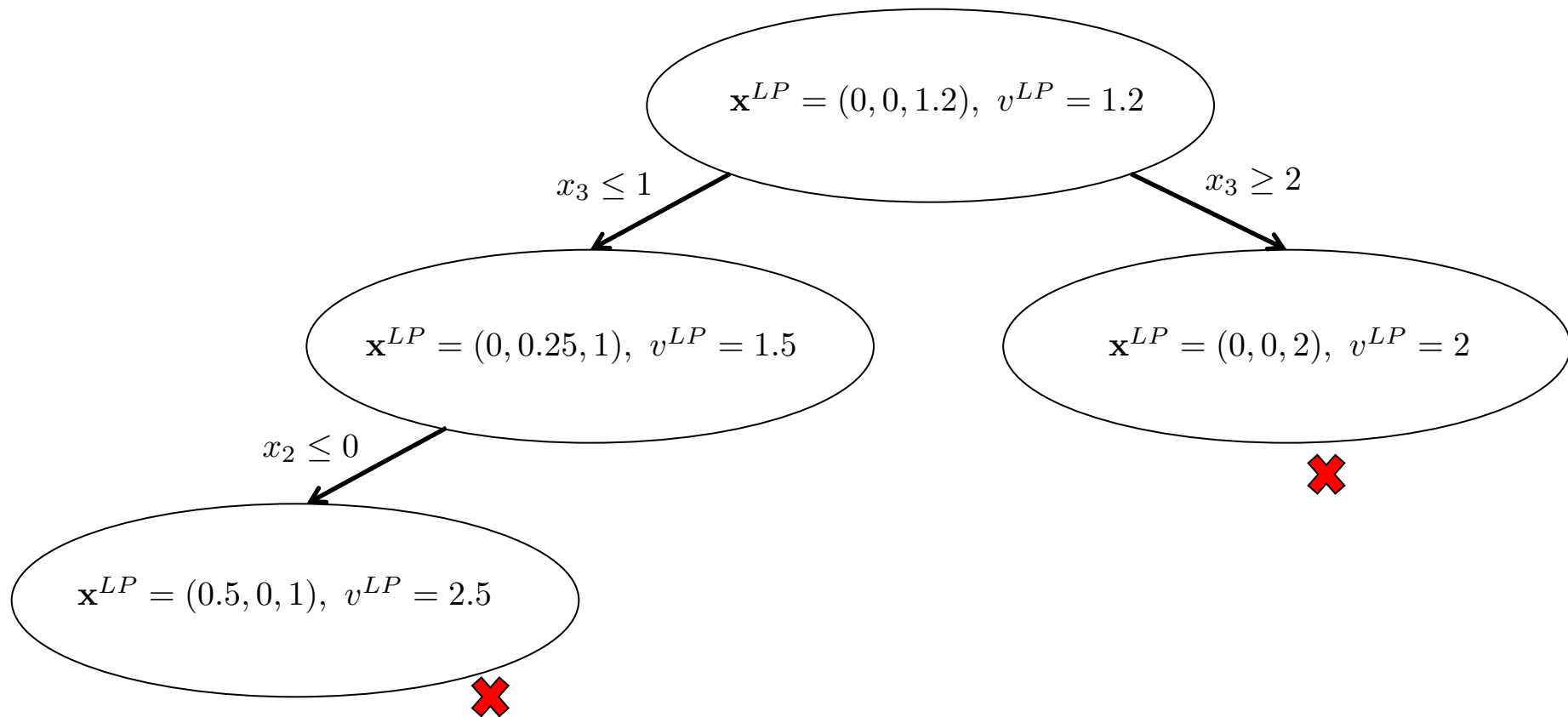
$$LB = 1.5 \leq v^{IP} \leq UB = 2$$

$$\begin{aligned} v^{LP} = \min \quad & 3x_1 + 2x_2 + x_3 \\ \text{s.t.} \quad & 3x_1 + 4x_2 + 5x_3 \geq 6 \\ & x_3 \geq 2 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$



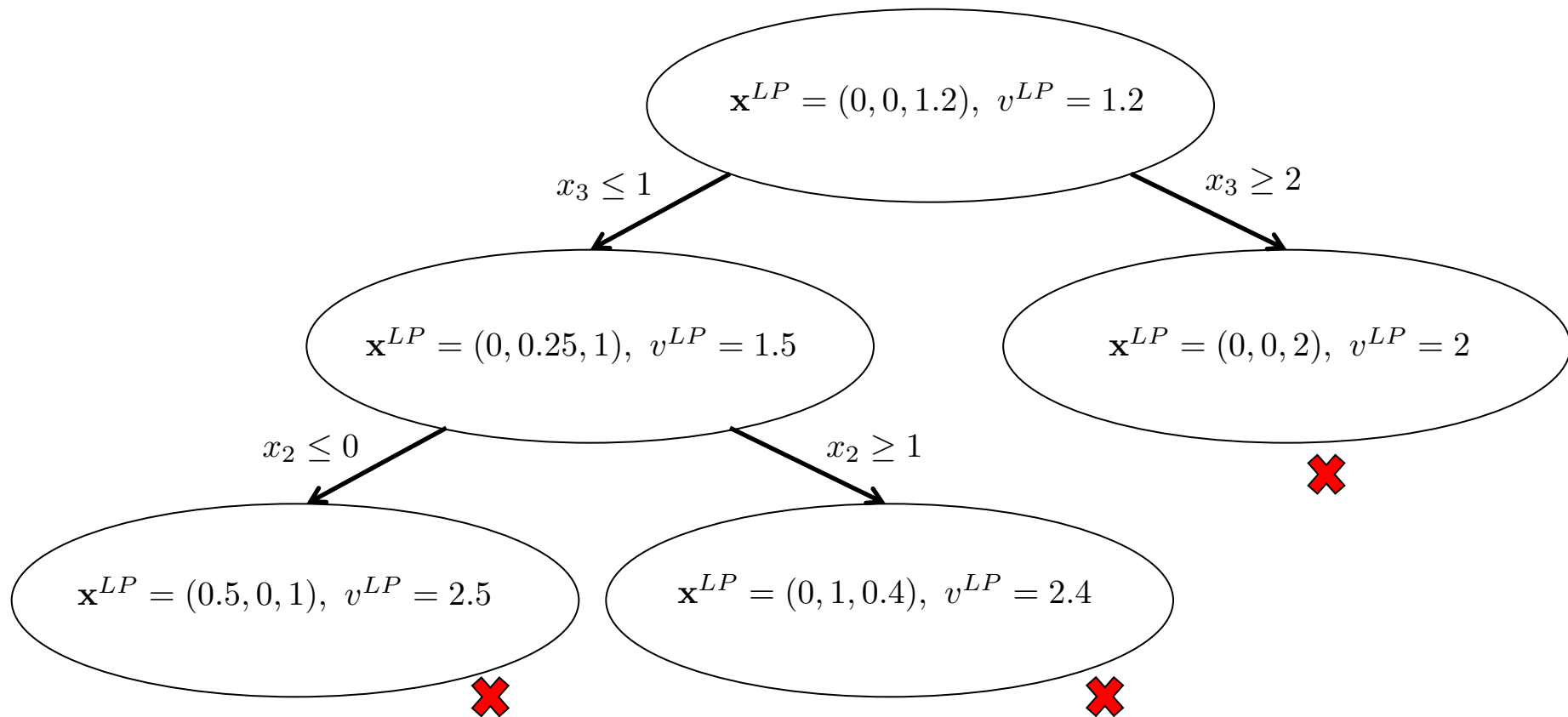
Branch-and-Bound

$$LB = 1.5 \leq v^{IP} \leq UB = 2$$



Branch-and-Bound

$$LB = 2 \leq v^{IP} \leq UB = 2$$



Summary

- We observed a simple example of branch and bound.
- Notice we have ties in selecting which node to process first and which variable we need to branch on.
- Sophisticated rules are used to make these choices.